# Utilizing Machine Learning with Tissue Specificity as Markers in Acute Myeloid Leukemia

Gautham Anne
Davuluri Lab

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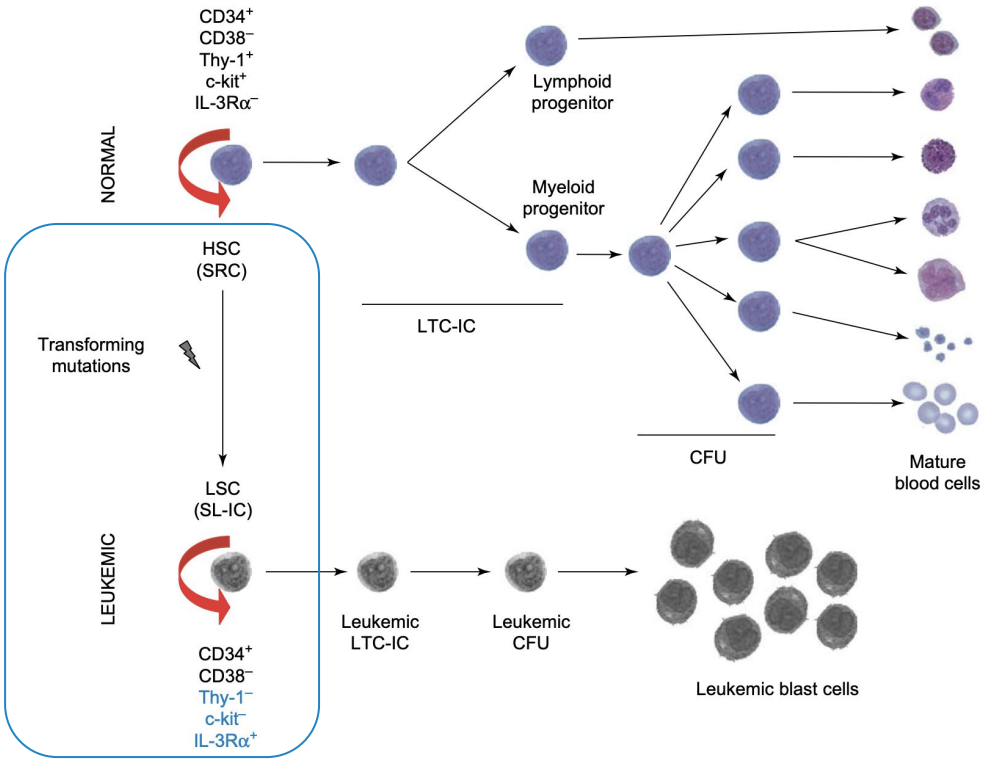
# Introduction




(Wang, J.C. et al., (2005))

# Goals

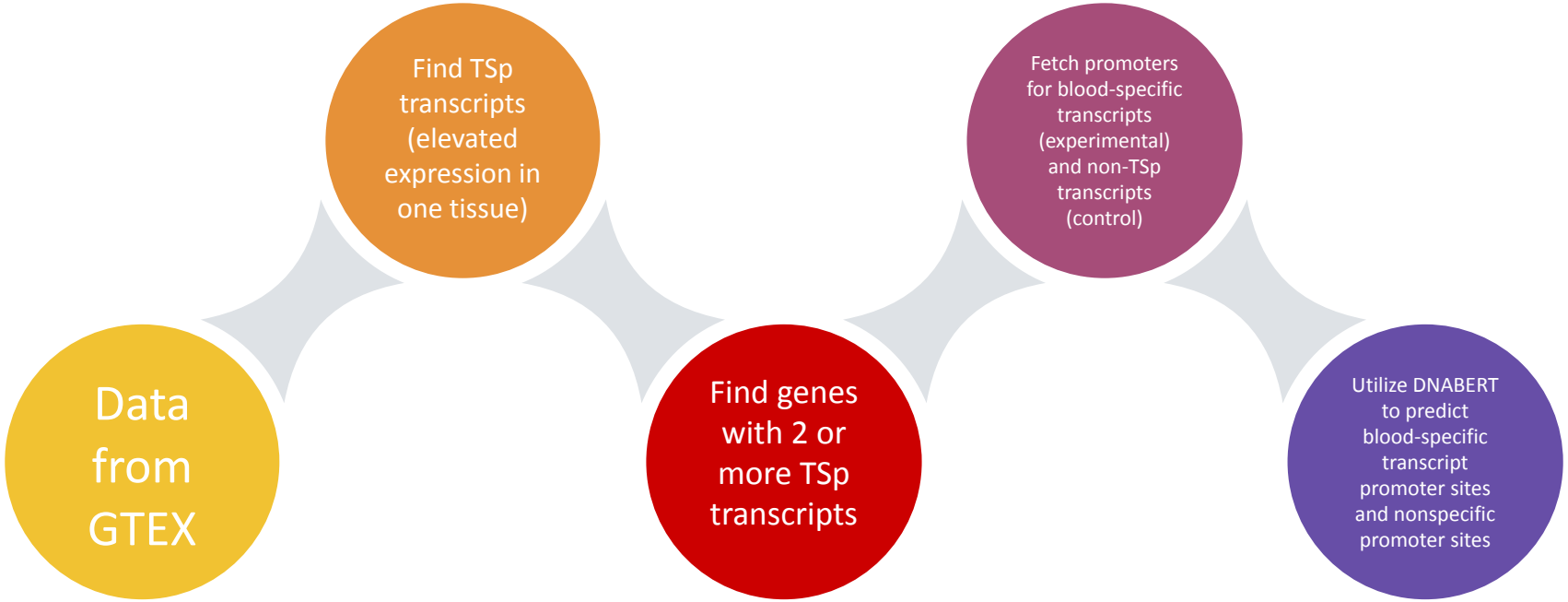
Use existing ML tools to understand biology in a deeper sense

Build DL/ML model to predict TSp promoters

DNABERT (DNA language model) to understand TSp promoters

Determine which transcript variants affect disease and how

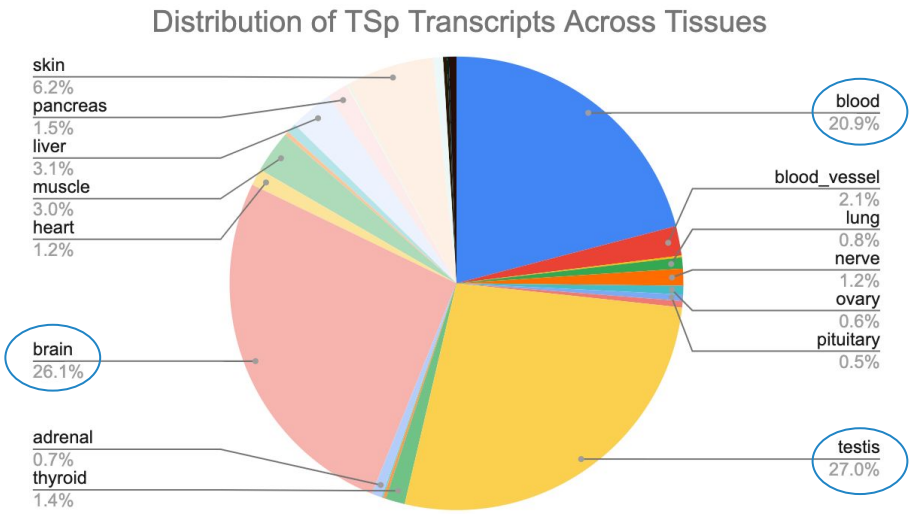
# Methods


Data from GTEX
Find TSp transcripts (elevated expression in one tissue)
Find genes with 2 or more TSp transcripts
Fetch promoters for blood-specific transcripts (experimental) and non-TSp transcripts (control)
Utilize DNABERT to predict blood-specific transcript promoter sites and nonspecific promoter sites

# Preliminary Results

- 6,641 genes with 2+ TSp transcripts

| | Upregulated | Upregulated in Blood |
|---|---|---|
| TSp Transcripts | 33,342 | 6,980 |
| Transcripts Present In More Than 10 Tissues | 72 | 18 |
| Unique Transcripts | 127,466 | 37,272 |




Results agreed with a recent paper (Zhu, J. et al., (2016))

# Thank you!

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